

Effects of coaching by community pharmacists on psychological symptoms of antidepressant users; a randomised controlled trial

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Abstract

Background: Community pharmacists strive to deliver pharmaceutical care to patients. At the moment, coaching of depressive primary care patients on taking their antidepressants (ADs) is not yet part of their standard care package. **Aims:** To investigate the effects of coaching by community pharmacists on psychological symptoms. **Method:** A randomised controlled trial with a 6-month follow-up. Outcomes: psychological symptoms with the Hopkins Symptom Checklist (SCL). Intention-to-treat (ITT) was performed with (1) last observation carried forward and (2) with group mean imputation (GMI). **Results:** Analyses with LOCF and GMI resulted in different findings. The LOCF method revealed that at the 6-month follow-up, the intervention patients were less depressed and less anxious than the controls. The intervention was particularly effective in patients with lower levels of education who received pharmacist's coaching. However, ITT with the GMI method showed no differences in psychological symptoms. Differences between LOCF and GMI were explained by the selective attrition in the intervention arm (attrition intervention patients had lower initial SCL-item scores on depression and anxiety than the completers) and by the higher attrition rate in controls. **Conclusions:** Our study indicates that the interpretation of the effects of an intervention on psychological symptoms can differ substantially by the way missing values are imputed. If both LOCF and GMI produce significant differences, efficacy can be concluded. If not, the effects based on ITT analyses with LOCF are based on artefacts. We recommend that positive intervention effects should only be reported when findings with LOCF and GMI are in accordance. © 2003 Elsevier B.V./ECNP. All rights reserved.

Keywords: Pharmaceutical care; Patient education; Pharmacist coaching; Depression

1. Introduction

Depression causes a tremendous burden for families, employers, and most of all for patients themselves. Major depression is the fourth leading cause of worldwide disease burden (George et al., 2000). Depressive symptoms and depressive disorders are among the most prevalent mental health problems. In the community population 1-year prevalence estimates vary between 2.2% and 5.8% for major depression (Angst, 1995; Zung et al., 1993; Bijl et al., 1998) and in primary care 1-month prevalence varies

between 11.1% (using DSM-III-criteria) and 13.5% (using ICD-10-criteria) (Tiemens et al., 1995). Global burden of depression will perhaps rank second only to that of ischaemic heart disease by the year 2020 (Murray and Lopez, 1996). Depression is characterised by repeated relapse, recurrent episodes (Paykel and Priest, 1992), chronic illness (Hirschfeld, 2001) and frequent comorbid anxiety (Ormel and Tiemens, 1995). Although randomised controlled trials leave little doubt that antidepressants (ADs) are effective medication against depressive disorders (Mynors-Wallis et al., 2000; Gill and Hatcher, 2002), lay people and many patients have a negative attitude towards ADs, which causes sub-optimal treatment effects (Priest et al., 1996; Demyttenaere, 2001). So far little or no attention has been paid to the pharmacy as a setting for interventions with ADs.

As in the USA (Smith and Benderev, 1991) and in

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Western European countries, community pharmacies in the Netherlands are moving from a product-centred towards a patient-centred model of practice, which is called pharmaceutical care. Dutch community pharmacies are implementing three new tasks, namely medication surveillance, informing prescribing physicians and patient medication education. A recent overview showed that the first two tasks are implemented in the pharmacist's daily routine activities, but patient education is not (Pronk et al., 2002). In the Netherlands most patients visit only one pharmacy, and very rarely switch or go 'shopping'. Therefore, pharmacists can easily monitor and coach patients on the (side) effects they can expect. Despite these advantages it is unclear whether patients can benefit from a pharmacist-coaching programme.

1.1. Objective

The objective of this article was to investigate the effects of coaching by community pharmacists and of an informative take-home video on psychological symptoms of depressed primary care patients who use antidepressants.

2. Methods

2.1. Design

A randomised controlled trial (RCT) with a follow-up at 3 and 6 months was conducted among consecutive patients who attended the pharmacy for non-tricyclic antidepressants. The trial consisted of two arms (usual care and pharmaceutical extra care) and patients were randomly assigned to one of the groups (see Fig. 1). Neither patients nor pharmacists were blinded for group assignment. The RCT tried to improve adherence and to reduce psychological symptoms by means of a coaching programme by community pharmacists. At baseline and at 3- and 6 month follow-up we measured presence and severity of psychological symptoms. The RCT had the approval of the medical-ethical committee of the University Medical Centre Utrecht.

2.2. Subjects and settings

From April 2000 to April 2001 a total of 19 pharmacists sought informed consent of consecutively attending patients. The enrolment period varied between 3 and 12

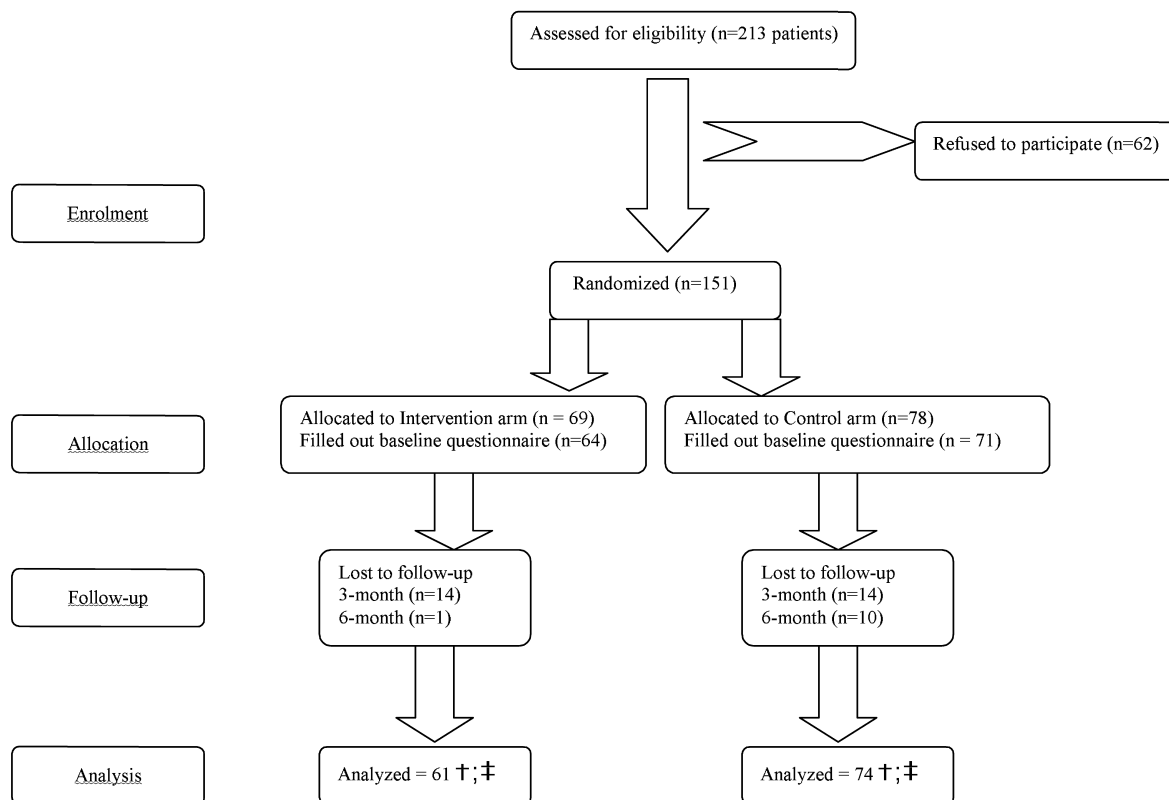


Fig. 1. Randomised controlled trial (RCT) (the CONSORT statement) (Moher D. et al., 2001). **Error!** Intention-to-treat: last observation data of 11 patients were carried forward. **Error!** Per protocol analysis was based on data of 103 patients.

months. Inclusion criteria for patients were: (1) 18 years old or over; (2) attending the pharmacy with a 'new episode' prescription for a non-tricyclic AD medication from their general practitioner (GP), i.e. not having used an antidepressant in a 6-month period before the inclusion. This was checked in the pharmacy records. The ADs comprised: Selective Serotonin Reuptake Inhibitors (SSRI), Reversible Inhibitor Mono-Amineoxidase (RIMA), Noradrenergic and Specific Serotonergic Antidepressant (NaSSA) and Serotonin Noradrenaline Reuptake Inhibitor (SNRI). The SSRIs included citalopram, fluoxetine, fluvoxamine, paroxetine, duloxetine, mianserine, and sertraline. The RIMA was moclobemide, the NaSSA mirtazapine and the SNRI venlafaxine; (3) the ability to understand and complete Dutch questionnaires, and (4) taking the antidepressants because of depressive complaints. The latter was checked by asking the patients whether the prescription was related to depressive symptoms.

Patients in the control group received the usual oral information when they collected their prescriptions at the pharmacy. Both intervention and control patients received the same informative printed material on ADs. The pharmacists asked written informed consent of patients.

The prescribing GPs were asked to complete and return a brief questionnaire to provide the indication(s) for prescription, its severity and possible co-morbidity. Study subjects were rewarded €12 for completing all questionnaires from baseline through follow-up. The pharmacists received €35 compensation for each patient they enrolled in the study and for coaching intervention patients over a period of 6 months.

The participants in both arms received their medication at the pharmacy and filled out and returned the self-rating questionnaires at baseline measurement and at 3- and 6-month follow-up. The pharmacists delivered two distinct service packages: usual care and pharmaceutical care (i.e. the intervention arm). Intervention patients were assigned to the pharmaceutical care group and had three coaching contacts during the study. In a few cases the assistant pharmacist took care of the contacts with the patients. At first contact they were informed about the appropriate use, the benefits and the side effects of the medication. The pharmacists were asked to use a list of important themes (Lin et al., 1995; Nierenberg, 1999) to discuss with the patients: (1) Take the medication daily, (2) take the ADs for 2–4 weeks for a noticeable effect, (3) continue to take the medicine even when feeling better, (4) do not stop taking ADs without checking with the physician and (5) do not hesitate to ask the pharmacist or the GP if you have questions regarding ADs.

To further improve their knowledge of ADs, the patients received a take-home video. The study team made the 25-min videotape, and it reflected the above points and also reviewed the multifactorial origin of depression, its

relationship to stress, physical and emotional symptoms, how medication and psychotherapy can relieve depression and the importance of medication adherence.

The second contact generally took place within 2 weeks before their first prescription ended, the third at 3 months from baseline. At the second visit the pharmacist asked the patient whether he/she had experienced adverse or positive effects. In both cases patients were stimulated and motivated to continue to take their ADs by emphasizing the importance for relapse prevention. If the patient had stopped using the medication, reasons for discontinuation were discussed and registered. At the 3-month follow-up all patients evaluated the contacts with the pharmacist as positive, although intervention patients were more positive.

2.3. Outcomes

The self-rating 90-items (Hopkins) Symptom Checklist (SCL-90) measured psychological symptoms (Lipman et al., 1979). Each item is scored on a 5-point scale: 1 = 'not at all'; 2 = 'a little bit'; 3 = 'moderately'; 4 = 'quite a bit'; 5 = 'extremely'. In our trial we found a high reliability of the 13-item depression subscale ($n=135$; $\alpha=0.88$). The following subscales also had high internal consistencies: the 10-item anxiety subscale ($n=126$; $\alpha=0.87$), the 10-item obsessive-compulsive subscale ($n=130$; $\alpha=0.83$), the 7-item phobic-anxiety subscale ($n=129$; $\alpha=0.79$), the interpersonal sensitivity subscale ($n=127$; $\alpha=0.81$) and the somatization subscale ($n=125$; $\alpha=0.83$). Depression was strongly associated with the following psychological symptoms: anxiety ($r=0.63$), phobic-anxiety ($r=0.45$), somatization ($r=0.52$), interpersonal sensitivity ($r=0.69$), and obsessive-compulsive symptoms ($r=0.74$), indicating high levels of psychological comorbidity.

We did not analyse the anger-hostility subscale ($\alpha=0.18$), the paranoid-ideation subscale ($\alpha=0.53$) or the psychoticism subscale ($\alpha=0.65$) because of their low reliability.

2.4. Sample size

Sample size was set at 150 subjects (75 patients per arm) based on the ability to detect a difference of 13% in depression at a significance level of 0.05 (two-sided), with 80% statistical power and assuming a standard deviation in depressive symptoms of 40%.

2.5. Randomization

Randomisation occurred at patient level and with a one to one ratio, using block randomisation to ensure equal numbers of intervention and control patients per pharmacy. In order to avoid crossover contamination of usual care with elements of the intervention during randomisation, all pharmacists attended a pre-trial meeting at which they

were instructed how to approach eligible patients, how to randomise them and how to use different protocols for control and intervention patients. The data administration forms of the whole sample were randomised before delivery to the pharmacies. These forms were pre-coded and delivered in sealed envelopes. After receiving written informed consent of the patient, the pharmacist learned the group assignment by opening the envelope.

2.6. Statistical methods

We performed χ^2 analyses on categorical variables, Spearman's correlation (including partial correlation) on ordinal variables, a non-parametric test for skewed variables (Mann–Whitney), independent sample *t*-tests including Levene's tests for equality of variances, univariate analyses of variance and repeated measures on continuous variables (SCL-subscales and quality of life) in a general linear model (GLM). GLM enables analysis of the effects of a repeated measure factor correcting for baseline scores. We explored the potential effect modification and confounding of demographic and disease-related variables on the effects of the intervention. These variables included age, gender, marital status, education, work situation, income, severity of depression as estimated by the GP, number of previous episodes of antidepressant use and/or of depressive complaints.

In our analysis we used the software programme SPSS-11 (SPSS, 2002).

2.7. Intention-to-treat and per protocol analyses

Furthermore, we used intention-to-treat (ITT) and per protocol analyses (PPA) analyses with (1) last observation carried forward (LOCF) and (2) group mean imputations (GMI) for both arms, separately at the 3- and 6-month follow-up. For the LOCF baseline data were imputed for missing data at 3-month follow-up ($n=28$) and 3-month data for 6-month follow-up ($n=11$), respectively. In order to measure the efficacy of the pharmacist coaching we also performed PPA using the following protocol criteria: 1. The AD prescription to both arms had to be written out by a GP; 2. Intervention patients indicated they had watched the videotape and 3. Intervention patients received three coaching contacts: at baseline, after 2 weeks (with a margin of 1 week) and after 12 weeks (a margin of 3 weeks earlier or later was allowed). Using the PPA the data set consisted of 103 patients (29 intervention and 74 control patients).

3. Results

3.1. Participant's flow and follow-up

Of the 46 pharmacists who were approached in 1999, 26

agreed to participate in the study. Of these 26 pharmacists, seven stopped shortly after the start of the study, two because of a long sick leave, two because they had no time after the departure of assistants, two became demotivated because several eligible patients refused and one because GPs refused to let their patients participate. The recruitment is summarized in Fig. 1.

The 19 pharmacists approached 213 patients and enrolled 151 patients. Immediately after randomisation four patients decided not to participate. One because he considered the questionnaire too long, one because she was admitted to a hospital because of heart problems, one did not start medication and one stopped because she disagreed with her GP's diagnosis of depression after seeing the video. This left 69 (47%) intervention and 78 (53%) control patients. At baseline 12 (8%) of the 147 patients who were initially included did not return their questionnaires, which resulted in a baseline data set of 135 patients. At the 3-month follow-up 107 (73%) (50 intervention and 57 control patients) returned their questionnaires. At the 6-month follow-up 96 (65%) (49 intervention and 47 control patients) returned their follow-up questionnaires.

3.2. Attrition

'Attrition' patients were defined as patients who did not send back their questionnaires at 3 and/or 6-month follow-up and 'completers' as those who completed the full 6-month trial and returned both questionnaires. Hence, the attrition rate at 3- and 6-month follow-up was 21% and 27%. The attrition rate was equally distributed between both arms at both follow-ups. There was an even distribution between the attrition patients and the completers on age, gender, marital status, education, life/work situation, income, number of previous episodes of antidepressant use and allocation to drug coaching or not. At baseline, completers reported having faith in the effectiveness of the ADs (71%) significantly more often than the attrition patients (29%) ($\chi^2=10.572$ df=1 $P=0.01$).

3.3. Baseline

At baseline measurement there were no significant differences between the intervention group ($n=64$) and control group ($n=71$) on any demographic or health-related variables, or on clinical symptoms (Table 1).

For the whole group the mean and median age was 42 years and 71% of the patients were female. The GPs made a primary diagnosis for 122 (90%) patients and a secondary diagnosis for 22 (16%). Of their primary diagnoses the GPs rated 19% as mildly, 65% moderately and 17% as severely disordered. There was an even distribution between both arms on both diagnoses; for 53 intervention patients and 56 control patients the GPs' primary diagnosis

Table 1

Demographic and clinical characteristics in the intervention (n=64) and control arm 22 (n=71) at baseline measurement

	Intervention	Controls	Degrees of freedom	Test	P value
Female	70%	69%	1	χ^2	0.770
Age (years) ^a	43±13	42±12	5	τ	0.898
<i>Severity of symptoms</i>			2	χ^2	0.987
Mild	18%	19%			
Moderate	66%	64%			
Severe	16%	17%			
<i>Marital status</i>			3	χ^2	0.152
Unmarried/single	19%	28%			
Married/living together	73%	56%			
Divorced	6%	9%			
Widowed	2%	7%			
<i>Life/work situation</i>			5	χ^2	0.407
Paid work	62%	66%			
Voluntary work	5%	—			
Domestic work	11%	15%			
Pensioner	8%	6%			
Unemployed/incapable of working	14%	13%			
<i>Monthly income</i>			5	χ^2	0.063
<\$650	36%	28%			
\$680–\$1000	31%	28%			
\$1000–\$1350	17%	24%			
\$1350–\$1650	9%	12%			
>\$1650	7%	8%			
<i>Level of education (diploma)</i>			5	χ^2	0.376
No education (not finished)	7%	1%			
Primary school	8%	10%			
10 years of education	27%	23%			
14 years of education	31%	39%			
16 years of education	27%	27%			
<i>Health status (in groups)^a</i>					
Months of depressive complaints	31±64.8	49±87.2	127	τ	0.169
Number of previous episodes use of AD ^b	35±0.86	23±0.49	131	τ	0.380
Number of previous episodes similar complaints	1.5±3.1	2.5±5.5	123	τ	0.197
<i>Clinical Characteristics^a</i>					
SCL-Depr. (total number) ^c	40±12	38±11	131	τ	0.160
(mean SL-CL-Depr.-items)	(3.2±0.9)	(2.9±0.8)			

^a Values are mean±S.D.^b AD=antidepressants.^c SCL Hopkins Symptom Checklist.

was depression and for 3 intervention and 10 control patients the primary diagnosis was anxiety disorder. As to the 22 secondary diagnoses, 4 intervention and 4 control patients were diagnosed with anxiety disorder by their GPs.

On average patients had experienced similar episodes of complaints twice before. The median for the duration of the current depressive episode was one year. In the 3

months before inclusion, patients had 2 contacts with their GP (S.D. 2.4) and 4 with other health professionals (S.D. 4.5). The mean item score of both patient arms on the depression subscale (i.e. a subscale from one to five) was 3.0 (S.D. 0.9). Although not significantly, intervention patients were slightly more depressed at baseline than the controls (3.2 versus 2.9) ($t=1.462$ $df=131$ $P=0.146$) (Table 1).

Table 2

Improvement on SCL-subscale (mean item scores) from baseline ($n = 135$) through 3–23 and 6-month follow-up of the intervention arm and the control arm; General Linear Model with Intention-To-Treat (ITT)^a using baseline data as LOCF^b ($n = 135$) and Per Protocol Analyses (PPA)^a ($n = 103$)

Subscales	ITT/ PPA ^a	Baseline		6-Month		<i>F</i> (DF = 1)	<i>P</i> Value ^c
		I	C	I	C		
Depression	ITT	3.1	2.8	1.9	2.1	4.987	0.027 ^c
	PPA	3.1	2.9	1.7	2.1	11.577	0.001 ^c
General	ITT	2.7	2.5	1.6	1.8	4.207	0.042 ^c
anxiety	PPA	2.6	2.5	1.5	1.8	9.610	0.003 ^c
Obsessive	ITT	2.8	2.7	1.9	2.1	2.487	0.117
compulsive	PPA	2.7	2.7	1.7	2.1	4.823	0.003 ^c
Phobic-	ITT	1.7	1.7	1.5	1.5	0.000	0.999
anxiety	PPA	1.7	1.7	1.2	1.8	2.657	0.107
Interpers.	ITT	2.2	2.2	1.7	1.7	0.338	0.562
sensitivity	PPA	2.2	2.2	1.4	1.8	5.256	0.024 ^c
Somatization	ITT	2.6	2.5	2.0	2.0	0.698	0.408
	PPA	2.5	2.5	1.7	2.0	5.530	0.021 ^c

^a ITT/PPA = Intention-To-Treat/Per Protocol Analyses.

^b LOCF = Last observation carried forward.

^c Improvement intervention arm is significant at the 0.05 level.

4. Improvement on psychological symptoms

4.1. Depression

At the 6-month follow-up more than two thirds of the patients were in remission (item score 1–2), 20% were slightly depressed (mean item score 2–3) and 11% had a mean item score of three or higher.

4.1.1. ITT and PPA with LOCF

Over a period of 6 months GLM analyses revealed that the intervention arm improved significantly more (baseline mean item score = 3.1 vs. 6-month item score = 1.9) than

the controls (baseline mean item score = 2.8 vs. 6-month item score = 2.1) on depressive symptoms ($F = 4.987$ df = 1 $P = 0.027$) (Table 2). Similarly, the per protocol analysis showed that intervention patients were less depressed (baseline mean item score = 3.1 vs. 6-month item score = 1.7) than the controls (baseline mean item score = 2.9 vs. 6-month item score = 2.1) ($F = 11.577$ df = 1 $P = 0.001$) (Table 2).

4.1.2. ITT and PPA with GMI

GLM analyses showed no differences between both arms on depression (Table 3).

Table 3

Improvement on SCL-subscale (mean item scores) from baseline ($n = 135$) through 3- and 6-month follow-up of the intervention arm and the control arm; General Linear Model with Intention-To-Treat (ITT)^a using group mean imputation (GMI)^b per arm separately ($n = 135$) and Per Protocol Analyses (PPA)^a ($n = 103$)

Subscales	ITT/ PPA ^a	Baseline		6-Month		<i>F</i> (DF = 1)	<i>P</i> Value ^c
		I	C	I	C		
Depression	ITT	3.1	2.8	1.8	1.8	0.819	0.4
	PPA	3.1	2.9	1.8	1.7	1.826	0.2
General	ITT	2.7	2.5	1.8	1.9	1.913	0.2
anxiety	PPA	2.6	2.5	1.7	1.8	3.420	0.07
Obsessive	ITT	2.8	2.7	1.8	1.9	0.713	0.4
compulsive	PPA	2.7	2.7	1.7	1.9	0.975	0.3
Phobic-	ITT	1.7	1.7	1.2	1.3	0.132	0.7
anxiety	PPA	1.7	1.7	1.2	1.3	0.977	0.3
Interpers.	ITT	2.2	2.2	1.5	1.6	0.194	0.7
sensitivity	PPA	2.2	2.2	1.4	1.6	0.226	0.6
Somatization	ITT	2.6	2.5	1.8	1.8	0.10	0.9
	PPA	2.5	2.5	1.7	1.8	0.744	0.4

^a ITT/PPA = Intention-to-treat/per protocol analyses.

^b GMI = Group mean imputation.

^c Improvement intervention arm is significant at the 0.05 level.

4.2. Anxiety

4.2.1. ITT and PPA with LOCF

Over the 6-month period GLM analyses revealed that the intervention arm improved significantly more (baseline mean item score=2.7 vs. 6-month item score=1.6) than the controls (baseline mean item score=2.5 vs. 6-month item score=1.8) on anxiety ($F=4.207$ $df=1$ $P=0.042$) (Table 2). Per protocol analyses showed that intervention patients (baseline mean item score=2.6 vs. 6-month item score=1.5) were significantly less anxious than the controls (baseline mean item score=2.5 vs. 6-month item score=1.8) ($F=9.610$ $df=1$ $P=0.003$) (Table 2).

4.3. Other subscales

4.3.1. ITT and PPA with LOCF and GMI

Per protocol analyses showed that the intervention patients improved significantly more on obsessive-compulsive symptoms ($F=4.823$ $df=1$ $P=0.003$), interpersonal sensitivity ($F=5.256$ $df=1$ $P=0.024$) and somatization ($F=5.530$ $df=1$ $P=0.021$) (Table 2) whereas no differences were found using GMI (Table 3).

4.4. Confounding factors and effect modification

The effects of the intervention on the psychological symptoms of the patients were not confounded by any demographic or disease-related variables that we introduced as GLM covariates. However, LOCF analyses showed that educational level was an effect modifier on depression ($F=8.255$ $df=1$ $P=0.005$). The intervention appeared to be more effective in patients with lower levels of education (less than 14 years of education) ($F=5.034$ $df=1$ $P=0.03$), which suggests that the effect of the educational coaching and the videotape is particularly important in patients with lower levels of education. By contrast, a high level of education (14 years of education or over) predicted a good outcome on depression in both arms. GMI demonstrated neither confounders nor effect modifiers.

4.5. LOCF and GMI

The differences between the LOCF and GMI analyses were explained by (1) the selective attrition in the intervention arm: attrition patients had initial lower SCL-item scores on depression and anxiety (mean item score 2.8 S.D. 0.9 and 2.2 S.D. 1.1, respectively) than the completers (mean item score 3.2 S.D. 0.8 and 2.7 S.D. 0.9, respectively) whereas within the control group the depression and anxiety item scores were comparable for dropouts and completers (mean depression item score 2.9 and anxiety item score 2.3); (2) the higher attrition rate in controls ($n=27$) compared to the intervention arm ($n=22$).

5. Conclusions

Our study indicates that the effects of an intervention on psychological symptoms may be biased by the way missing values are replaced, i.e. by LOCF and GMI, and may lead to different clinical outcomes. ITT with LOCF led to significantly less depressive and anxious symptoms in the intervention patients, especially in patients with lower levels of education. And thirdly, per protocol analyses revealed that the intervention patients improved more on obsessive-compulsive symptoms, interpersonal sensitivity and somatization. However, with the GMI method no effects on psychological symptoms were found.

We recommend that positive intervention effects should only be reported when findings with LOCF and GMI are in accordance. If not, the findings after LOCF are artefacts. We are aware that our study has some limitations. We do not know the impact of the video compared to the pharmacist contacts and which elements of these two components contributed most to the decrease of depression and anxiety. As to the contacts, the pharmacists were asked to use a list of important themes to discuss with the patients. Yet, we did not measure the quality of each of these contacts or whether the information was conveyed correctly. In addition, the video was made for patients with depressive complaints. For some patients who had a primary diagnosis of anxiety, the video may have fulfilled their educational needs to a lesser degree. Although informing prescribing physicians about medication takes place in a structural way in the Netherlands through drug therapy meetings between groups of GPs and pharmacists (Pronk et al., 2002), there is no structural exchange of information between both disciplines about medication at the individual patient level, and it is rare. A possible obstacle might be that GPs are afraid their autonomy might be encroached upon (Porsius, 2001). However, we are optimistic about cooperation and information exchange between both disciplines in the near future, because most of the GPs approached to participate in the study allowed their patients to be coached by pharmacists, and provided additional patient data. Furthermore, 90–95% of the pharmacists and GPs nowadays take part in drug therapy meetings between both groups. The goal of these meetings is to reach an agreement on the rational prescription of medication (Kocken and Etienne, 1998).

The intensity of the contacts may vary over time and per patient, the busy schedule of the pharmacist makes it difficult to make follow-up appointments and last but not least there is the hectic daily routine at the pharmacy that may hamper an intimate, confidential conversation. According to the pharmacists themselves their assistants can take over some of the contacts with some additional training. The latter are involved in most of the contacts and have made a middle level professional education and are licensed to dispense prescription drugs under the supervision of the pharmacist (Blom, 1996). Pharmacy assistants

receive training in communication skills in their 3-year professional education. Communication is trained weekly through the whole curriculum, totaling at least 240 student hours. In the first year, the emphasis is on basic principles for verbal and nonverbal communication. In the second year a module in 'conversation' is taught and in the third year an integration of different subjects is taught in a so-called 'practice course'. Whether it is the pharmacist or the assistant, in our view it is important that in the first contacts educational and motivational messages are given that are aimed at correcting the misconceptions of patients that will most likely adversely impact adherence (Lin et al., 1995; Nierenberg, 1999).

In conclusion, both LOCF and GMI must be taken into account in replacing missing values assessing possible intervention effects. If both LOCF and GMI produce significant differences, efficacy may be concluded. If not, the conclusions based on ITT analyses with LOCF are based on artefacts and lead to wrong conclusions.

6. Declaration of interest

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